

FRANCISTOWN, Botswana

WMO: 680540

Lat: 21.160S

Lon: 27.475E

Elev: 1001

StdP: 89.87

Time Zone: 2.00 (E02)

Period: 96-15

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	1.9	3.5	-8.9	2.0	10.2	-5.6	2.6	12.0	9.4	17.4	8.4	18.0	1.4	20	0.398

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11	13.1	36.8	19.3	35.4	19.3	34.2	19.2	23.3	29.3	22.8	28.7	22.3	28.3	4.1	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.0	18.8	25.3	21.2	17.9	24.5	21.0	17.7	24.2	75.1	29.2	72.9	29.0	71.1	28.0	27.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.1	8.2	7.4	DB	-1.9	38.7	2.2	1.2	-3.5	39.5	-4.7	40.2	-6.0	40.9	-7.6	41.8	(4)
(5)			WB	-4.0	24.8	1.7	1.0	-5.2	25.5	-6.2	26.1	-7.2	26.7	-8.4	27.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	21.3	25.1	25.0	23.8	20.9	17.5	15.1	14.4	17.6	21.6	24.6	25.5	24.8	(6)	
	DBStd	4.71	1.81	1.83	2.22	2.35	2.53	2.61	2.04	3.02	2.92	2.99	2.96	2.44	(7)	
	HDD10.0	2	0	0	0	0	1	1	0	0	0	0	0	0	(8)	
	HDD18.3	335	0	0	1	5	46	102	122	51	7	2	1	1	(9)	
	CDD10.0	4127	468	421	426	327	233	153	137	234	349	454	465	458	(10)	
	CDD18.3	1421	210	187	169	82	21	4	1	27	106	198	216	201	(11)	
	CDH23.3	16306	1845	1769	1509	928	638	272	214	718	1650	2471	2494	1798	(12)	
(13)	CDH26.7	7273	768	772	628	334	181	56	29	247	827	1313	1307	810	(13)	
(14)	Wind	WSAvg	3.3	3.4	3.4	3.2	3.1	2.6	2.7	2.8	3.2	3.7	4.3	3.8	3.5	(14)
(15) Precipitation	PrecAvg	453	96	82	43	27	6	3	0	1	8	29	68	85	(15)	
	PrecMax	670	364	348	161	94	76	35	5	14	60	126	165	293	(16)	
	PrecMin	116	1	2	0	0	0	0	0	0	0	0	11	3	(17)	
	PrecStd	146	84	73	46	24	14	9	1	3	15	33	39	69	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	35.2	35.9	35.9	34.1	32.2	30.1	29.2	33.2	36.8	38.2	38.3	35.9	(19)	
		MCWB	20.8	20.5	19.4	19.2	16.4	15.6	14.6	16.6	18.2	19.5	20.0	20.7	(20)	
	2%	DB	33.8	34.1	34.0	32.1	30.2	28.3	27.3	31.2	35.0	36.8	36.5	34.2	(21)	
		MCWB	20.8	20.1	20.3	18.5	16.4	15.4	13.7	15.5	17.2	18.9	19.7	20.4	(22)	
	5%	DB	32.2	32.9	32.2	30.5	28.9	26.4	25.9	29.4	33.3	35.1	34.9	33.0	(23)	
		MCWB	20.9	20.6	20.4	18.4	16.0	14.3	13.3	15.2	16.7	18.5	19.4	20.7	(24)	
	10%	DB	30.9	31.1	30.4	28.9	27.1	24.8	24.1	27.8	31.9	33.6	33.2	31.2	(25)	
		MCWB	21.2	20.4	20.2	18.1	15.3	14.0	13.0	14.6	16.5	18.3	19.5	20.5	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	24.2	23.6	23.2	22.6	19.7	18.4	16.3	18.2	21.0	22.0	22.8	23.8	(27)	
		MCDB	29.9	29.4	29.7	27.9	27.5	25.5	26.6	29.4	31.0	32.9	32.6	30.5	(28)	
	2%	WB	23.4	22.9	22.5	21.5	18.2	16.7	15.0	17.0	19.3	21.0	21.9	22.9	(29)	
		MCDB	28.7	28.7	28.9	26.9	27.4	24.7	24.9	28.5	30.0	32.4	30.5	29.4	(30)	
	5%	WB	22.8	22.4	21.9	20.5	17.2	15.7	14.2	16.2	18.3	20.2	21.3	22.3	(31)	
		MCDB	27.9	28.1	28.0	26.2	26.2	23.5	23.3	27.0	29.4	31.5	29.6	28.6	(32)	
	10%	WB	22.3	21.8	21.4	19.5	16.2	14.8	13.4	15.4	17.4	19.4	20.7	21.7	(33)	
		MCDB	27.3	27.4	27.0	25.5	24.4	22.7	22.3	25.6	28.5	30.0	28.8	27.7	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	10.8	11.6	11.7	14.0	18.5	18.2	18.6	18.4	17.0	14.6	13.1	10.8	(35)	
		MCDBR	13.8	15.0	15.4	18.3	21.1	21.8	23.1	21.5	20.9	18.3	16.5	14.7	(36)	
	5% WB	MCWBR	4.8	5.2	5.6	7.8	10.0	11.5	12.5	10.4	9.0	6.8	5.7	4.8	(37)	
		MCWBR	10.2	10.7	11.4	12.6	17.4	18.1	19.3	18.7	17.7	15.0	13.0	10.9	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.365	0.362	0.349	0.321	0.289	0.285	0.286	0.337	0.417	0.416	0.385	0.373	(40)	
		taud	2.456	2.460	2.492	2.523	2.568	2.547	2.531	2.350	2.153	2.200	2.347	2.429	(41)	
	Ebn at Noon		979	969	957	944	941	925	935	913	876	907	953	972	(42)	
		Edh at Noon	121	118	110	99	87	85	89	115	152	152	134	124	(43)	
(44) All-Sky Solar Radiation	RadAvg		6.67	6.36	5.90	5.34	4.95	4.34	4.59	5.48	6.23	6.72	6.61	6.36	(44)	
	RadStd		0.59	0.72	0.58	0.39	0.16	0.29	0.19	0.21	0.31	0.38	0.50	0.56	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page